Exercises: Day 2

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Exercises

https://github.com/Applied-Ontology-Education/World-s-Ontology-Ecosystem/tree/main/exercises/day-2

Design Pattern Guidance

- Include a legend or key
- Visually distinguish **classes** from **instances** from **strings**
- Distingish **direct relations** from **shortcuts**
- Read arrows as reflecting "any instance of class X arrow some instance of class Y"
- Carefully determine how specific/broad to make your design patterns

OWL Encoding Guidance

- Read off the classes, instances, and relations from the design pattern
- Add axioms to classes using the "subclass of" button in Protege
- Add axioms to individuals under the "Type" button in Protege
- Use string literals for text values and xsd for datatypes as appropriate

Case 1

In aircraft_data.xlsx you will find a row for the Airbus A320 Neo.

Construct a design pattern and OWL encoding reflecting the content of columns C-G, R-S, and AB.

Case 2

In aircraft_data.xlsx you will find a row for the Airbus A321-111, designed to have a maximum knot approach speed of 142. However, after 5 approaches, an instance has obtained an average maximum knot approach speed of 139.

Construct a design pattern and OWL encoding reflecting the preceding phenomena.

Case 3

In soc_structure_definitions.xlsx you will find three "SOC_TITLE" entries that mention "Aerospace Engineer".

Construct a design pattern and OWL encoding that reflects all three entries and their respective "SOC Definitions".

Case 4

In employment_wage_May_2024.xlsx you will find three "OCC_TITLE" entries that mention "Aerospace Engineer".

Construct a design pattern and OWL encoding that reflects all three entries and their associated columns A, H, I, K, L.